

Bhushan Ladgaonkar

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PROFESSIONAL SUMMARY

Security Researcher and Machine Learning Engineer specializing in **Adversarial AI** and **Offensive Security**. Proven experience in engineering ML-based Network Intrusion Detection Systems (NIDS) for enterprise infrastructure and researching autonomous decision-making agents using Reinforcement Learning. Seeking to leverage expertise in simulation, modeling, and security architectures.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering
Bachelor of Technology (B.Tech) in Computer Engineering

Mumbai, India
Expected May 2028

EXPERIENCE

Oil and Natural Gas Corporation (ONGC)

Summer Intern | Database & Security Group

Mumbai, India
May 2025 – June 2025

- Developed a prototype **Network Intrusion Detection System (NIDS)** using the UNSW-NB15 dataset to identify zero-day threats and malicious traffic patterns, addressing specific limitations in traditional signature-based detection.
- Optimized data pipelines for the **UNSW-NB15 dataset** (2M+ records) using **SMOTE** to handle class imbalance, performing advanced feature engineering to enhance model accuracy for enterprise-grade network traffic.
- Implemented and evaluated **Random Forest** and **XGBoost** algorithms, successfully reducing false positive rates while maintaining high detection sensitivity for anomalous packets.
- Collaborated with the Database Group to document threat detection methodologies, bridging the gap between raw data analysis and actionable security intelligence.

TECHNICAL PROJECTS

TISD-Edu-AI: Local-First RAG Tutor | *Python, MLX, ChromaDB, FastAPI, LLMs*

Mar 2026

- Engineered a privacy-focused, on-device **Retrieval-Augmented Generation (RAG)** pipeline using **Microsoft Phi-3 (3.8B)** and **ChromaDB** to ground AI responses strictly in verified academic texts, eliminating educational hallucinations.
- Optimized hardware inference for Apple Silicon via the native **MLX framework**, achieving a high generation speed of **41.42 tokens/sec** and low latency (1.72s) while maintaining a minimal 9.99 GB memory footprint.
- Ingested and processed 4,600+ pages of academic PDFs into 6,400+ semantic chunks, validating system reliability with a custom evaluation suite that achieved **81.1% semantic accuracy** across a 100-sample stress test.

MendikotZero: AI Agent for Complex Trick-Taking Card Game | *Python, PyTorch, RL*

Oct 2025

- Architected a Reinforcement Learning agent capable of strategic decision-making in imperfect-information environments using the **AlphaZero architecture**.
- Implemented **Monte Carlo Tree Search (MCTS)** guided by a Dual-Head Neural Network (Policy & Value) to simulate future game states and optimize move selection.
- Designed a **Self-Play** training loop where the agent evolved strategies from random noise to advanced probability-based gameplay without human data input.

TECHNICAL SKILLS

Languages: Python, C, SQL, Java

AI & ML Systems: Machine Learning, Reinforcement Learning, RAG, LLMs, MCTS, PyTorch, MLX, ChromaDB, XGBoost

Security: Network Intrusion Detection (NIDS), Offensive Security, OSINT, Linux (Kali/Ubuntu), Reconnaissance

Developer Tools: Git, GitHub, FastAPI, VS Code, Jupyter Notebooks

CERTIFICATIONS

CS50 Cybersecurity – Harvard University

Offensive Security Operations – Cybrary

Open Source Intelligence (OSINT) – Cybrary

Cyber Kill Chain – Cybrary